

Moulay Ismail University
National School of Arts and Crafts

Department of Industrial Engineering and Production
Master's Thesis Defense

Project Title :

**Community Detection in Social Networks for Targeted Marketing and
Recommender Systems**

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Academic Year: 2025 - 2026



Presentation Outline

01 Introduction & Problem Statement

02 Background & Key Concepts

03 Literature Review

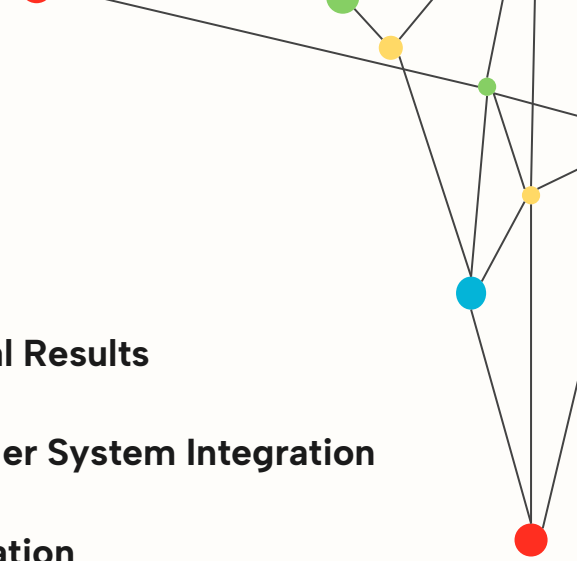
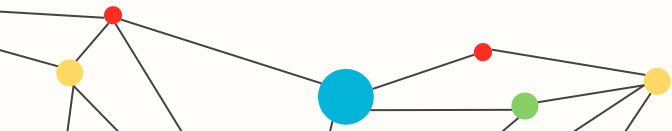
04 Methodology

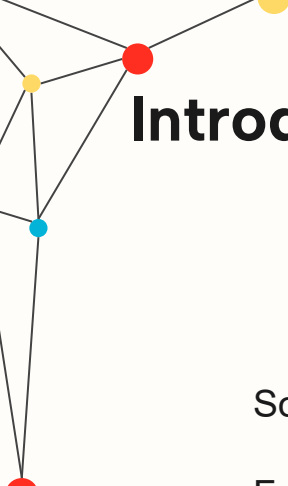
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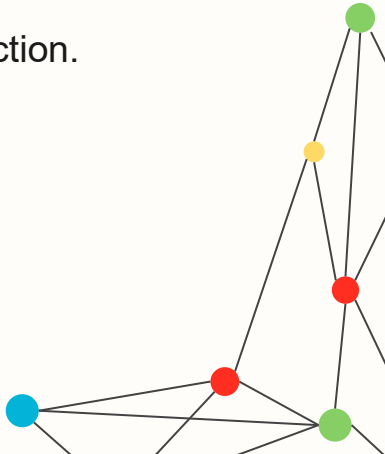
Introduction

Social media platforms generate massive networks of user interactions.

Facebook: 2+ billion active users.

Interactions include likes, shares, comments, follows.

These interactions form intricate networks revealing patterns of human connection.





Problem Statement

Despite numerous community detection algorithms available, selecting the right one remains challenging:

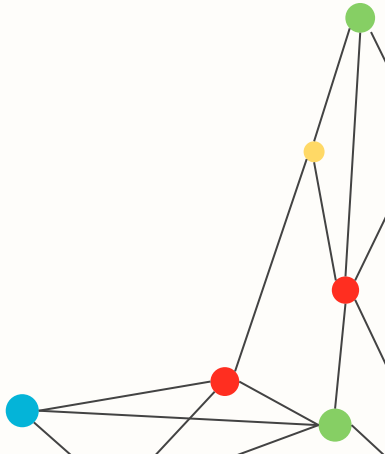
1. **Algorithmic Diversity** Different principles (modularity optimization, random walks, deep learning).
2. **Evaluation Gap** Most studies use modularity that don't predict practical usefulness.
3. **Limited Comparison** Modern vs. classical methods not systematically compared, especially at scale.
4. **Practical Utility** Lack of unified evaluation framework for real-world applications.

Research Question: *How do different community detection approaches compare in effectiveness and practical utility for recommender systems?*



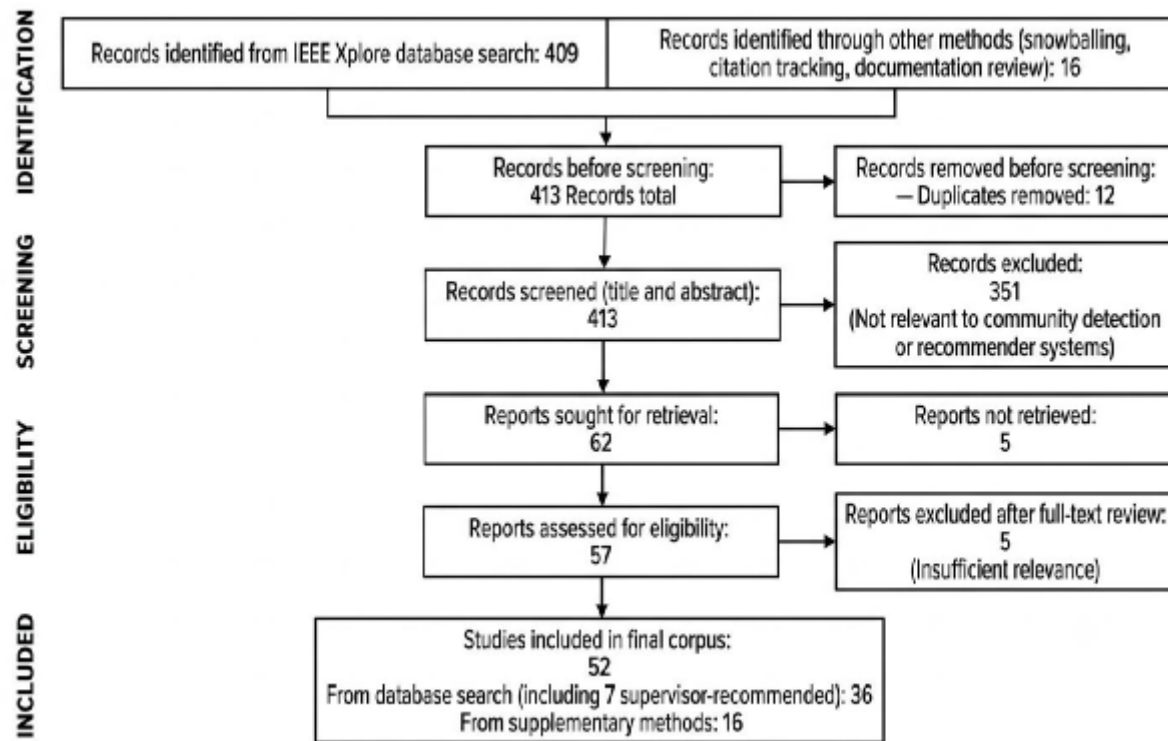


Research Objectives

1. Study and categorize community detection approaches (classical & modern)
 2. Implement representative algorithms from different categories
 3. Apply algorithms to real-world and synthetic network datasets
 4. Evaluate using structural metrics (modularity, NMI, ARI)
 5. Assess practical usefulness for targeted marketing & recommendation
 6. Compare performance and identify strengths/limitations
 7. Provide insights on trade-offs between simplicity, scalability, and effectiveness
 8. Build an interactive web application for illustration.
- 

Literature Review – Methodology

Prisma flow diagram



Literature Review – Classical Algorithms

Modularity-Based Approaches

<u>Algorithm</u>	<u>What It Does</u>	<u>Limitation</u>
Louvain (Blondel et al.)	Greedy modularity optimization.	Traag et al.: up to 25% communities poorly connected.
Leiden (Traag et al., 2019)	Guarantees well-connected communities.	Sliwa et al.: detected 97 vs. 41 communities vs. Louvain.

Random Walk-Based Approaches

<u>Algorithm</u>	<u>What It Does</u>	<u>Limitation</u>
Walktrap (Pons & Latapy)	Random walks trapped in dense regions.	Gates et al.: strong on flow-based networks.
Infomap (Rosvall & Bergstrom).	Minimizes map equation.	Yang et al.: linear time, top performer
MCL (van Dongen)	Expansion + inflation operators.	Papadopoulos et al.: pre-weighting improves NMI 0.4→0.8.



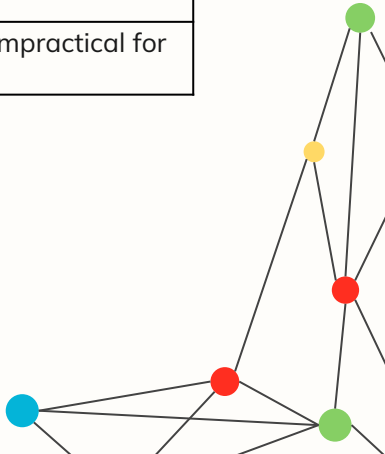
Literature Review – Classical Algorithms

Divisive Approaches

<u>Algorithm</u>	<u>What It Does</u>	<u>Limitation</u>
Girvan-Newman	Removes highest betweenness edges	Gates et al.: $O(n^3)$, impractical beyond few thousand nodes.
Kernighan-Lin (1970)	Greedy bisection with local search	Li et al.: requires equal-sized partitions.

Statistical Physics-Based Algorithms

<u>Algorithm</u>	<u>What It Does</u>	<u>Limitation</u>
SpinGlass (Reichardt & Bornholdt)	Physics-inspired optimization with simulated annealing	Yang et al.: high complexity; impractical for networks > 10,000 nodes



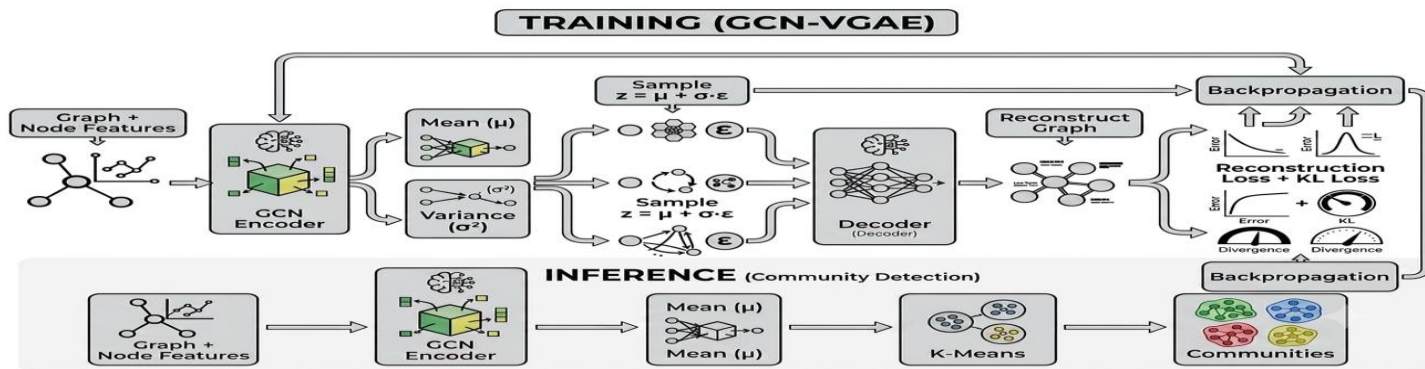
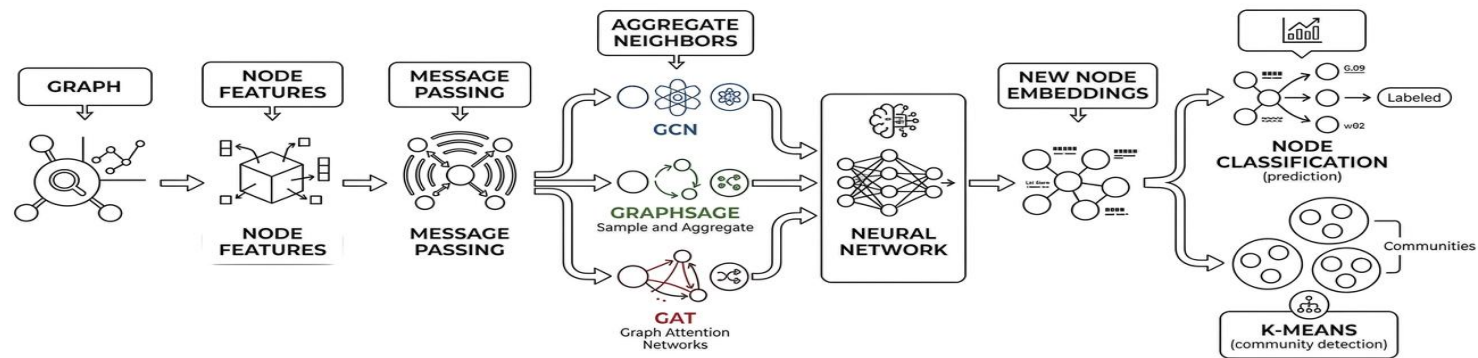
Literature Review – Classical Algorithms

Label Propagation Approaches

<u>Algorithm</u>	<u>What It Does</u>	<u>Limitation</u>
LPA (Raghavan et al.)	Nodes adopt most frequent neighbor label.	Gates et al.: fundamentally unstable.
FLPA	Asynchronous updates + memory	Yang et al.: outperforms LPA for $\mu \in [0.5, 0.7]$.
LPA-MNI (Li et al.)	Modularity + node importance.	Enhanced accuracy and robustness.
CLPA (Li et al.)	Constrained optimization.	Versatile for different network types.
FluidC (Parés et al.)	Communities as expanding fluids.	Anuar et al.: ARI ~0.9 on 500K nodes.

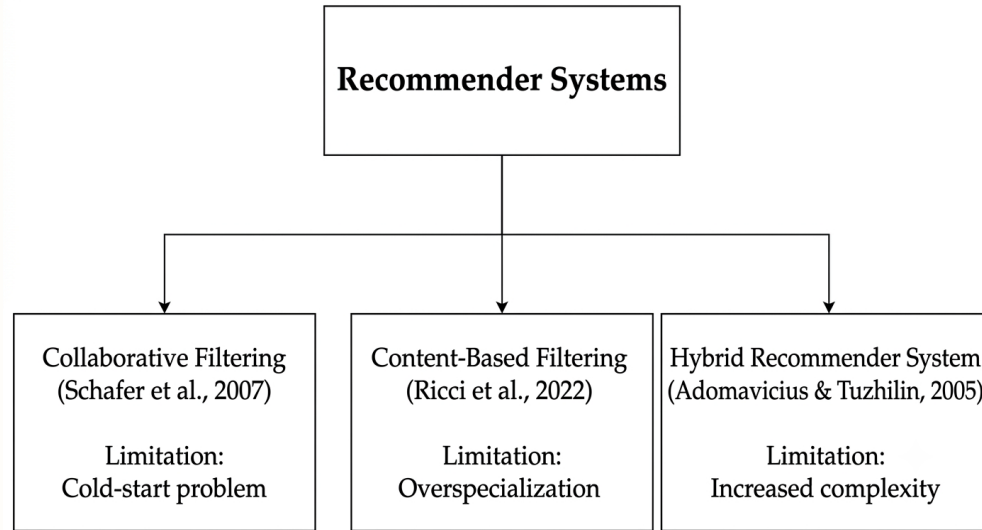
Literature Review - Modern Approaches

Graph Neural Networks



Literature Review – Recommender Systems

Three Classical Approaches



Challenges : Data Sparsity – Cold Start – Scalability – Interpretability.

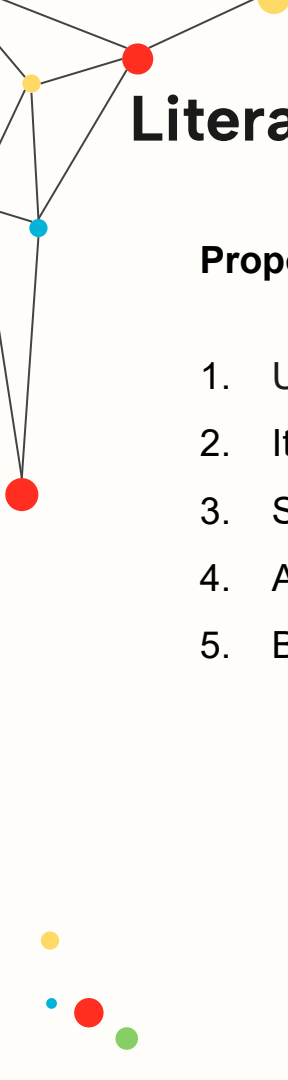
Literature Review – Applications

Targeted Marketing and E-Commerce

<u>Study</u>	<u>Application</u>	<u>Results</u>
Paul et al. (2025)	Commerce Graph.	Customers with high centrality enable personalized promotions.
Nasirzonouzi (2025)	Customer segmentation.	Reveals complex interrelations K-means fails to capture.
Kanavos et al. (2024)	Twitter community analysis.	Community structures influence information dissemination.

Recommender Systems Applications

<u>Study</u>	<u>Application</u>	<u>Results</u>
Saikalyan et al. (2024)	Local community detection	Effective cold-start on MovieLens and Facebook
Hu et al. (2024)	GGCRF framework	Addresses long-tail effect in recommendation
S.K.S. et al. (2025)	Textual semantics + GNNs	Improves community detection accuracy



Literature Review – Applications

Proposed Hybrid Approaches

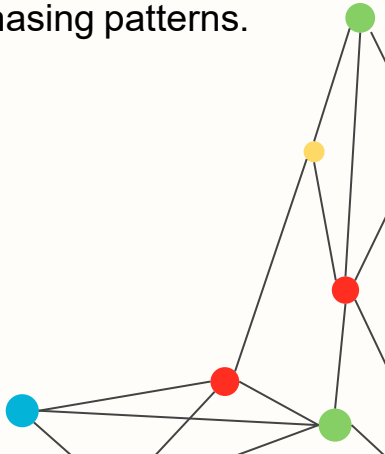
1. User-Based
2. Item-Based
3. SVD + Community
4. ALS + Community
5. BM25 + Community

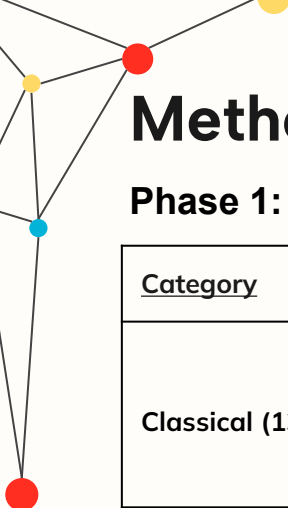
How They work

Community structure → Constrains candidate space, reduces sparsity.

Latent factors (SVD/ALS) → Captures fine-grained preferences.

BM25 → Leverages statistical co-purchasing patterns.





Methodology Overview

Phase 1: Community Detection Evaluation

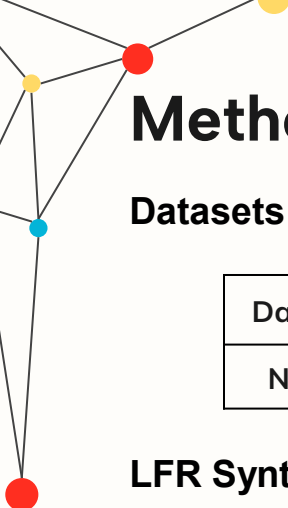
<u>Category</u>	<u>Algorithms</u>	<u>Datasets</u>	<u>Metrics</u>
Classical (13)	Louvain, Leiden, Walktrap, Infomap, LPA, FLPA, LPA-MNI, CLPA, FluidC, Girvan-Newman, Kernighan-Lin, SpinGlass, MCL	9 real-world + LFR synthetic (6 sizes)	NMI, ARI, Modularity
Deep Learning (4)	GCN, GAT, VGAE, Autoencoder	9 real-world + LFR synthetic	NMI, ARI, Modularity

Phase 2: Recommender System Integration

- Recommendation Approaches: User-Based, Item-Based, SVD+Community, ALS+Community, BM25+Community.
- Dataset: UCI Online Retail (541,909 transactions)
- Primary Metric: F1@5

Phase 3: Web Application





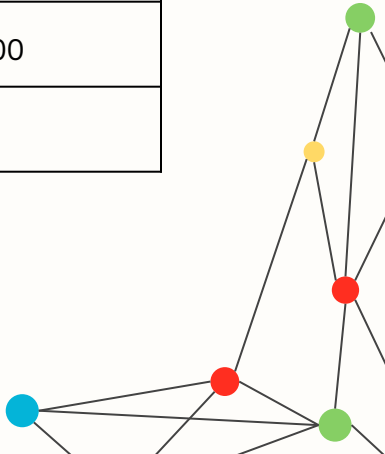
Methodology Overview – Datasets Used

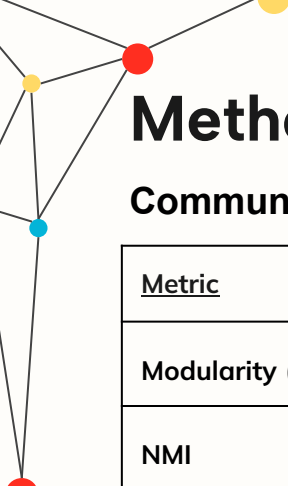
Datasets Summary

Dataset	Karate Club	Dolphins	Football	Polbooks	Primary School	Facebook	LastFM
Nodes	34	62	115	105	236	4,039	28,281

LFR Synthetic Networks

Parameter	Values
Network Size (n)	500, 1,000, 5,000, 10,000, 50,000, 100,000
Mixing Parameter (μ)	0.3, 0.5, 0.7





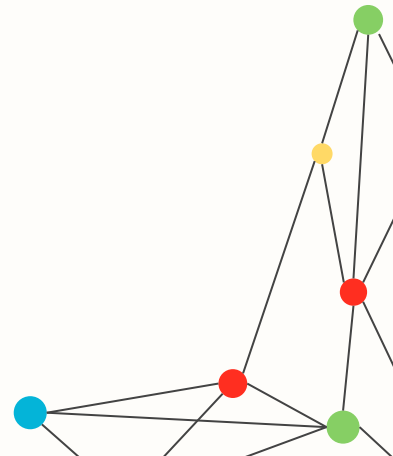
Methodology Overview – Evaluation Metrics

Community Detection Metrics

<u>Metric</u>	<u>Purpose</u>	<u>Range</u>
Modularity (Q)	Quality without ground truth	[-0.5, 1.0]
NMI	Similarity to ground truth	[0, 1]
ARI	Similarity adjusted for chance	[-1, 1]

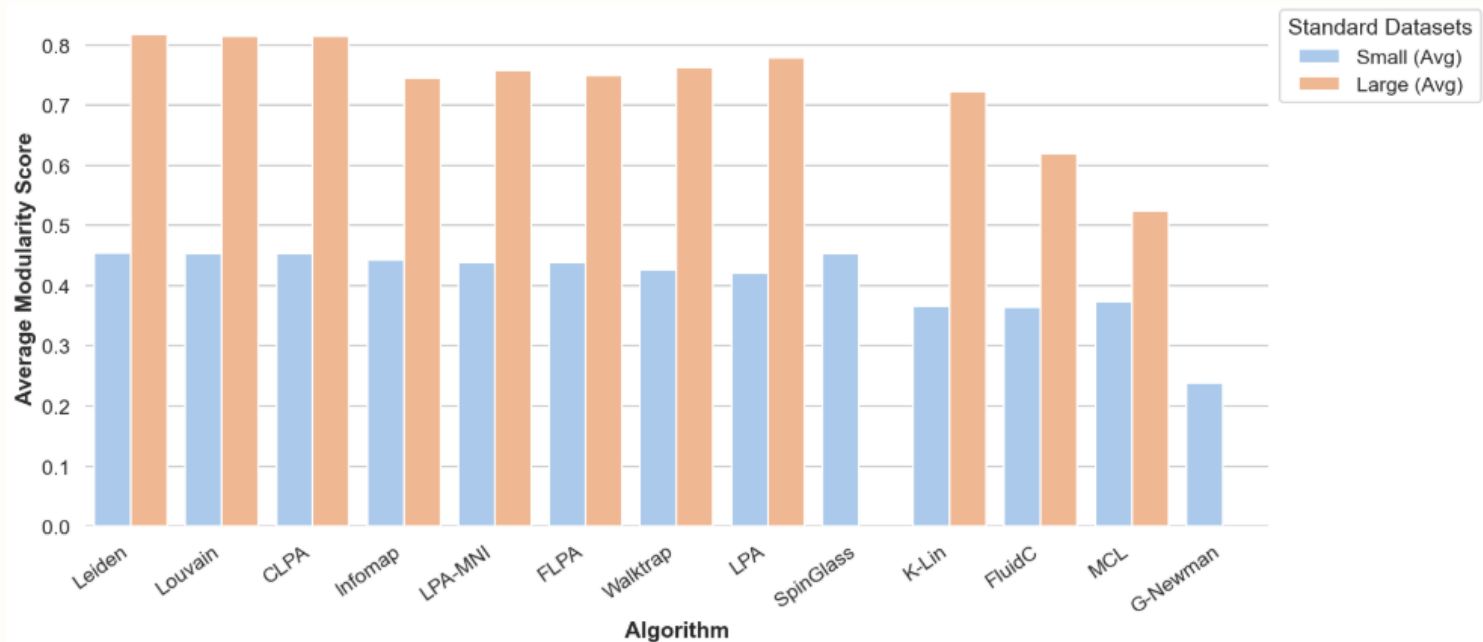
Recommender System Metrics (@K)

- Precision@K : Relevant items in top-K / K
 - Recall@K : Relevant in top-K / Total relevant
 - F1@K : $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$
- 



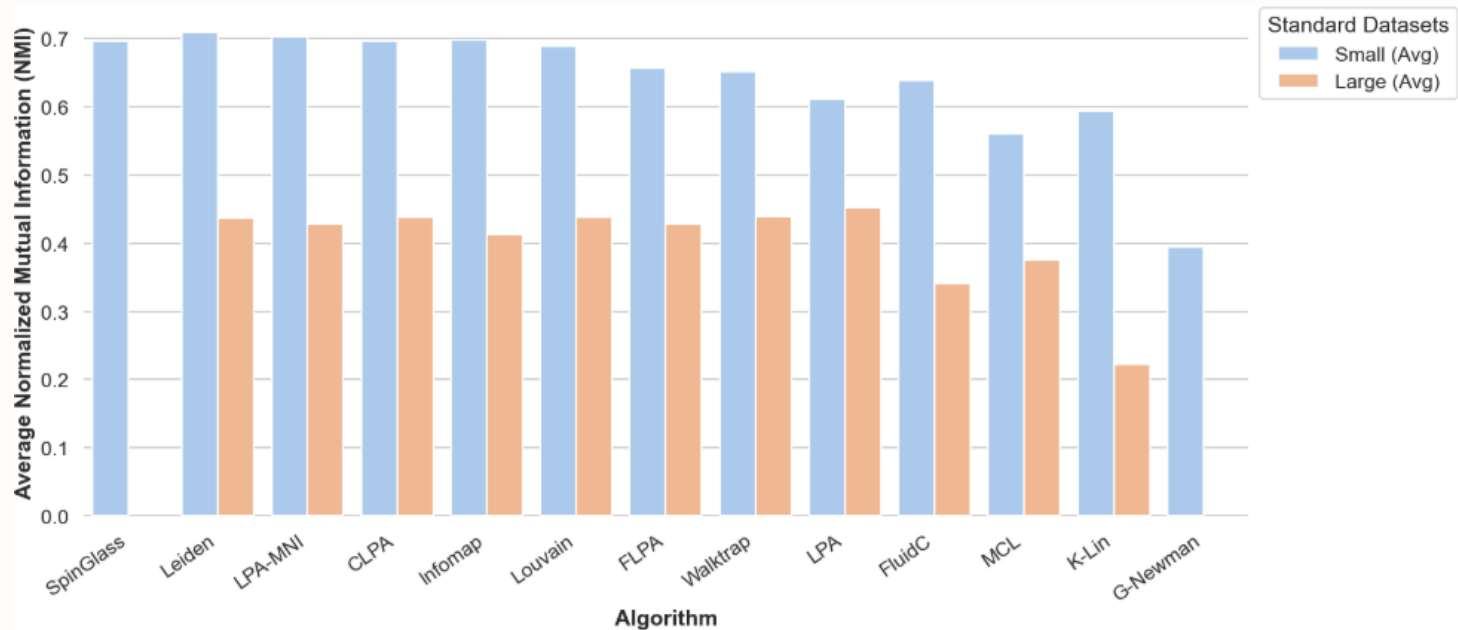
Results – Classical Algorithms

Community Detection Metrics - Modularity



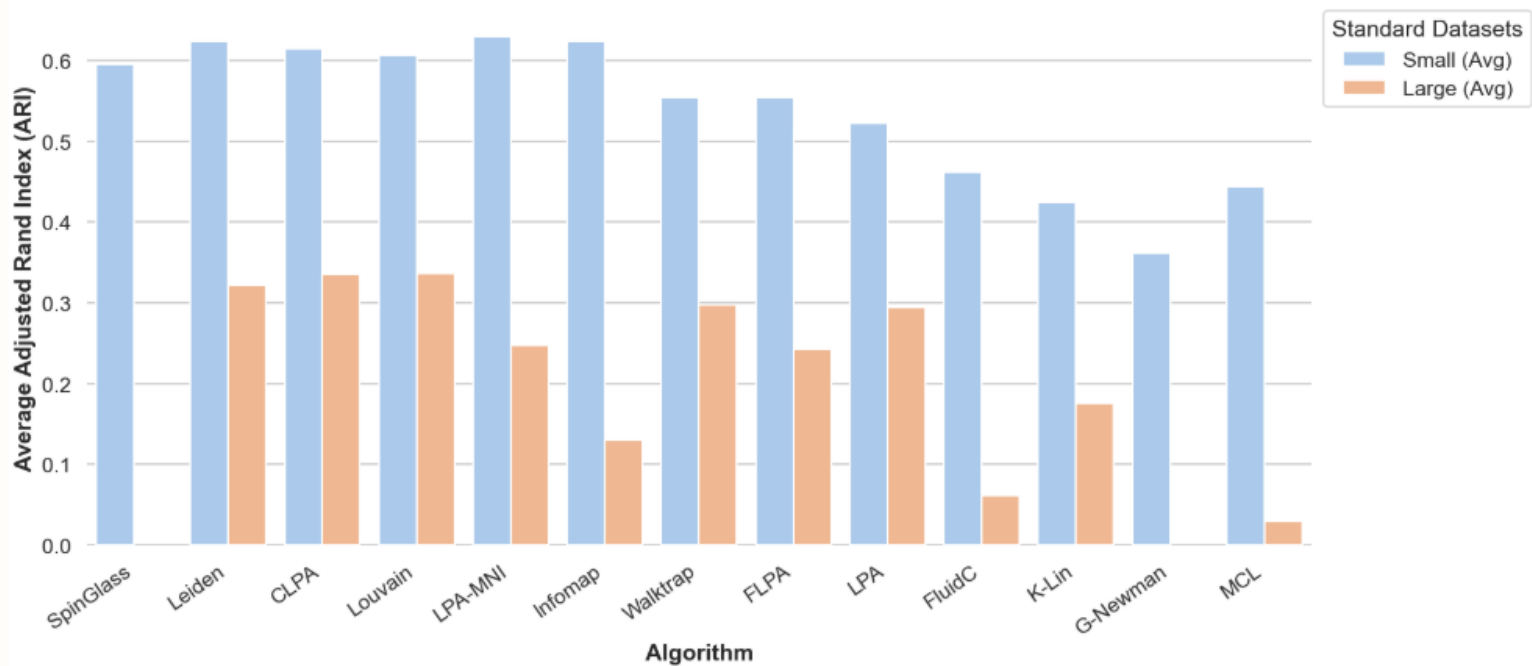
Results – Classical Algorithms

Community Detection Metrics - NMI



Results – Classical Algorithms

Community Detection Metrics - ARI



Results – Deep Learning Methods

Results - Deep Learning Methods - NMI

Algorithm	Karate	Dolphins	Football	Polbooks	PS	VS13	VS15	FB	LastFM
GCN	0.8372	0.8888	0.925	0.5745	0.1421	0.7871	0.6342	0.0486	0.0313
GAT	0.5756	0.8165	0.9195	0.4592	0.1119	0.8214	0.6343	0.0441	0.0295
VGAE	0.7324	1	0.9231	0.445	0.1263	0.7886	0.6095	0.0348	0.0258
Autoencoder	0.093	0.0279	0.8973	0.1635	0.1177	0.6207	0.567	0.0525	0.0296

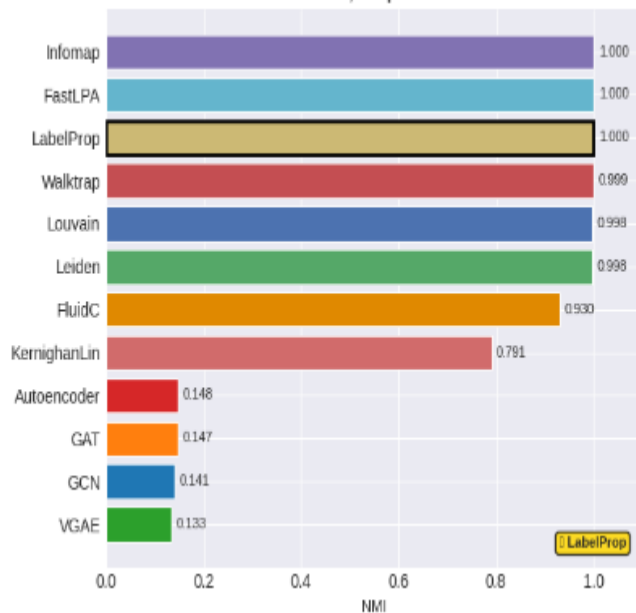
Results - Deep Learning Methods - ARI

Algorithm	Karate	Dolphins	Football	Polbooks	PS	VS13	VS15	FB	LastFM
GCN	0.8823	0.9348	0.8995	0.6745	0.0272	0.6808	0.4339	0.0614	0.0073
GAT	0.6682	0.8725	0.8955	0.4903	0.0129	0.7574	0.4490	0.0528	0.0064
VGAE	0.7717	1.0000	0.8905	0.4953	0.0234	0.7252	0.4248	0.0278	0.0049
Autoencoder	0.0072	0.0578	0.8358	0.0181	0.0144	0.3707	0.3719	0.0622	0.0061

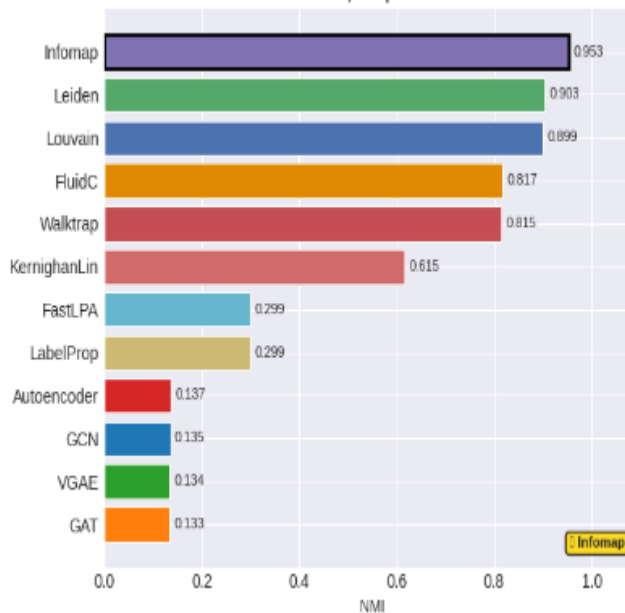
Results – LFR Benchmark

Community Detection Metrics - NMI

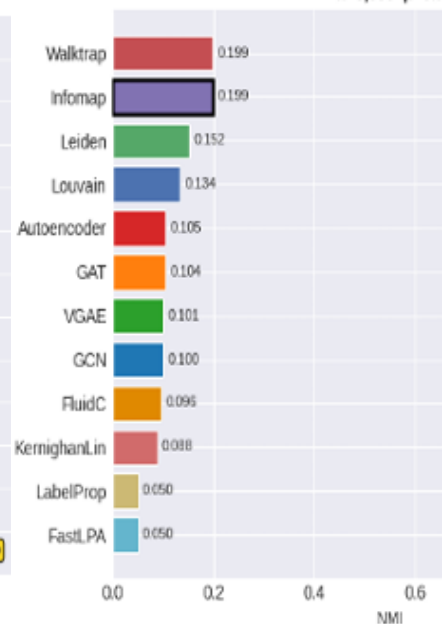
n=5,000 $\mu=0.3$



n=5,000 $\mu=0.5$

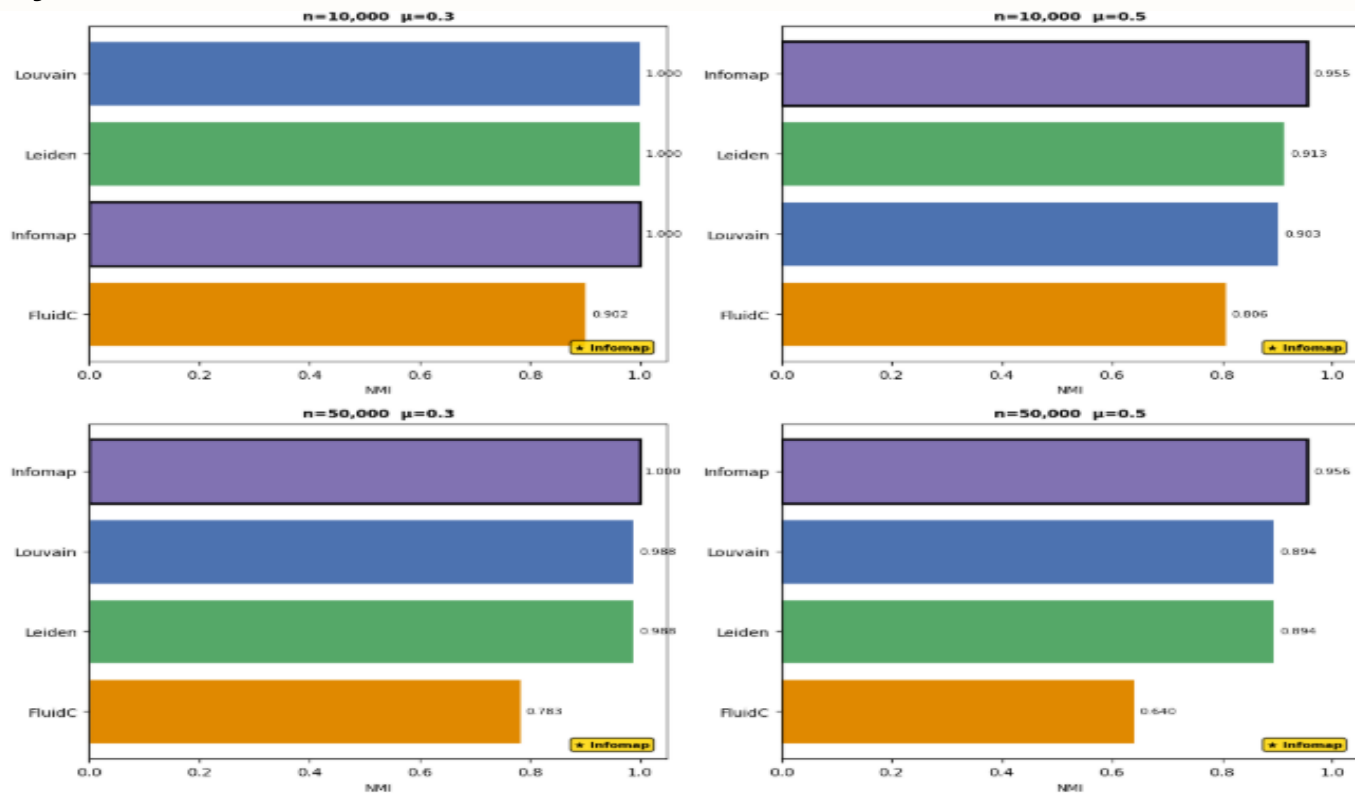


n=5,000 $\mu=0.7$



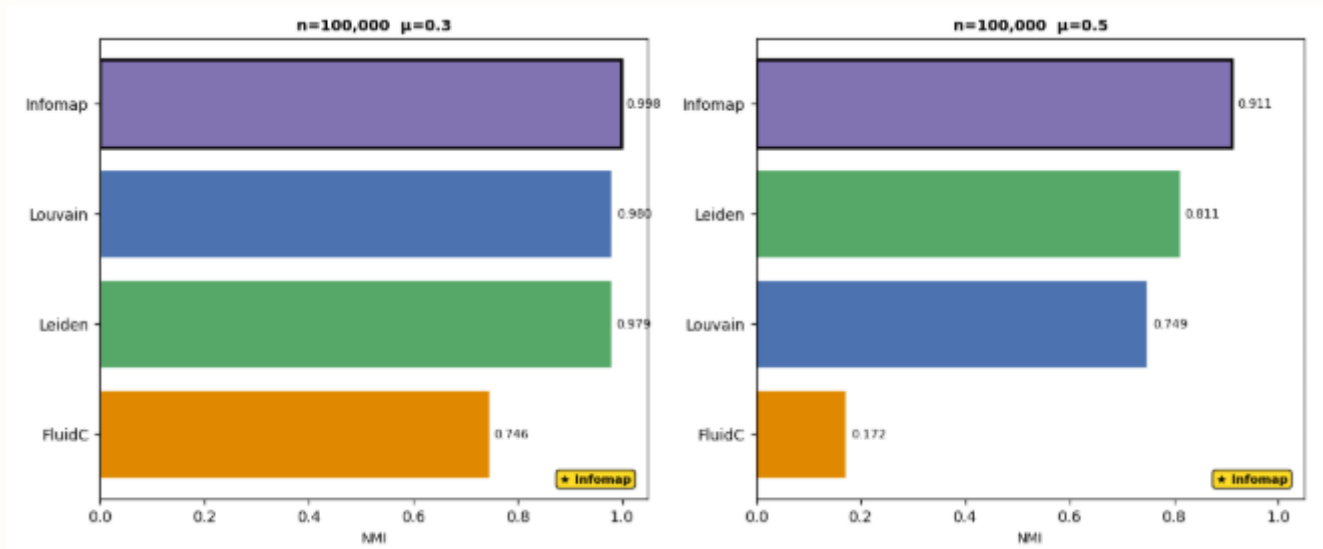
Results – LFR Benchmark

Community Detection Metrics - NMI



Results – LFR Benchmark

Community Detection Metrics - NMI



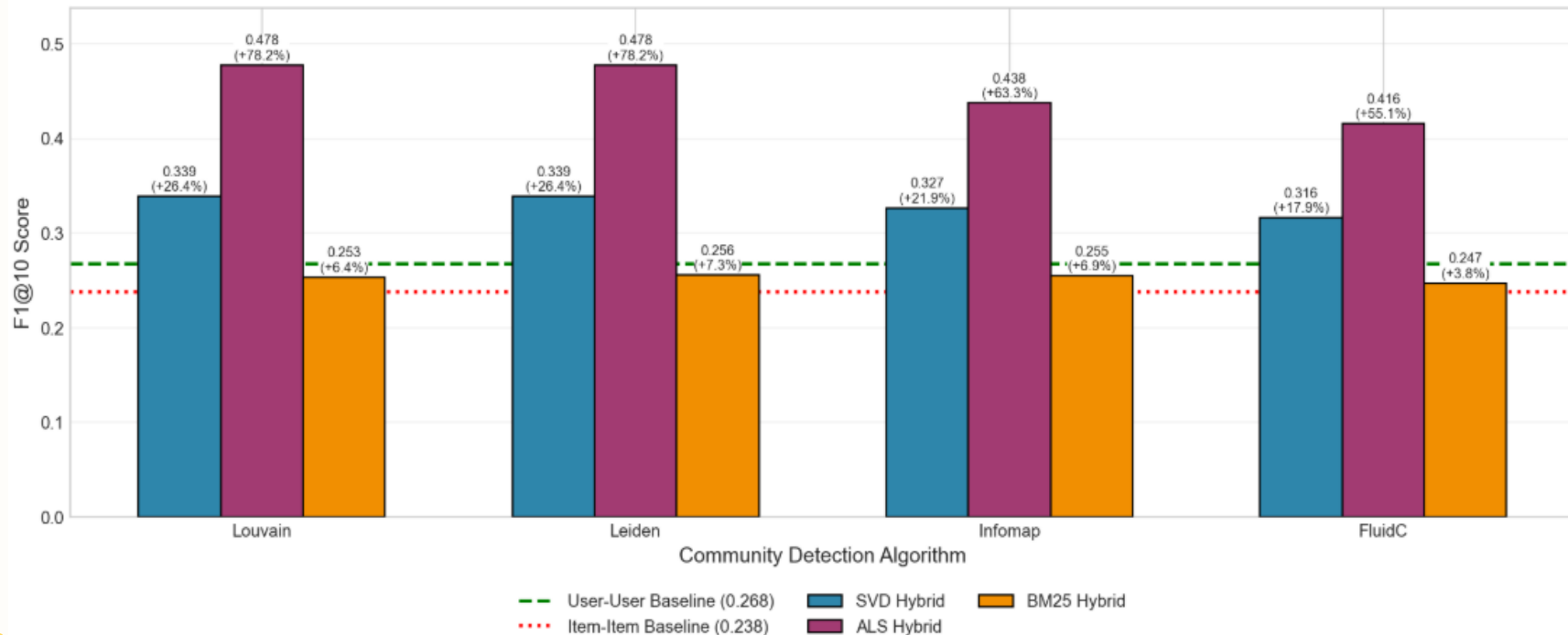
Results – Recommender System Integration

Hyperparameter Tuning Results

<u>Model</u>	<u>Default Config</u>	<u>Default F1@5</u>	<u>Tuned Config</u>	<u>Tuned F1@5</u>
ALS	factors=20, reg=0.1, $\alpha=40$	0.149	factors=100, reg=0.1, $\alpha=10$	0.371
BM25	k1=1.5, b=0.75	0.178	k1=3.0, b=1.0	0.211
SVD	n_components=50	0.277	n_components=50	0.277

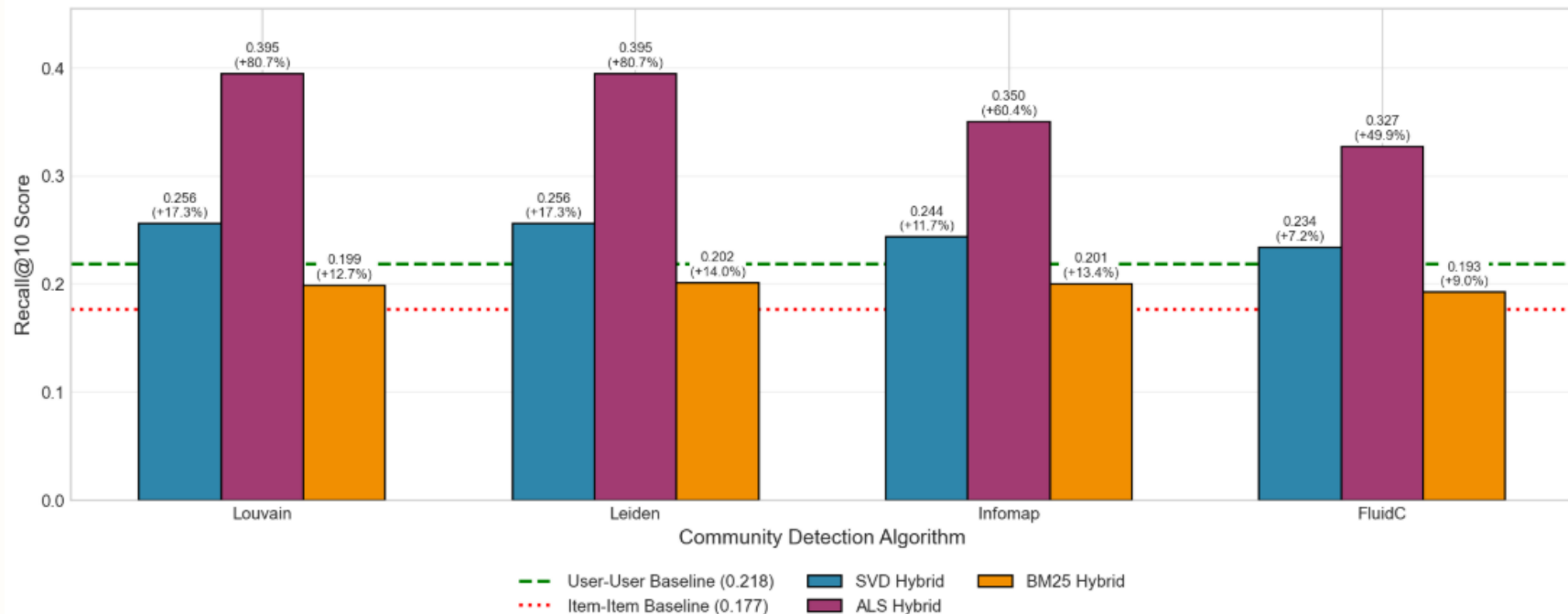
Results – Recommender System Integration

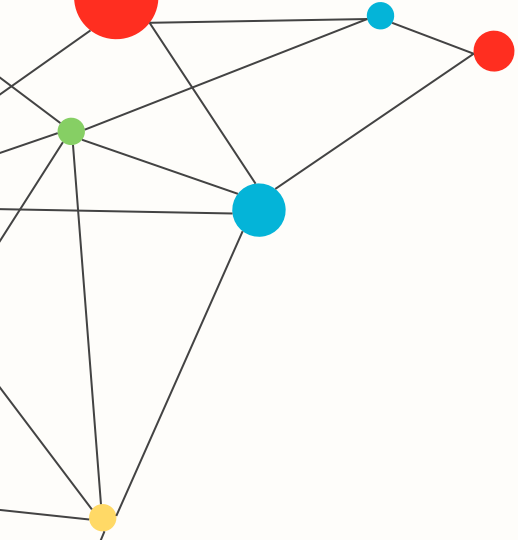
Performance Summary



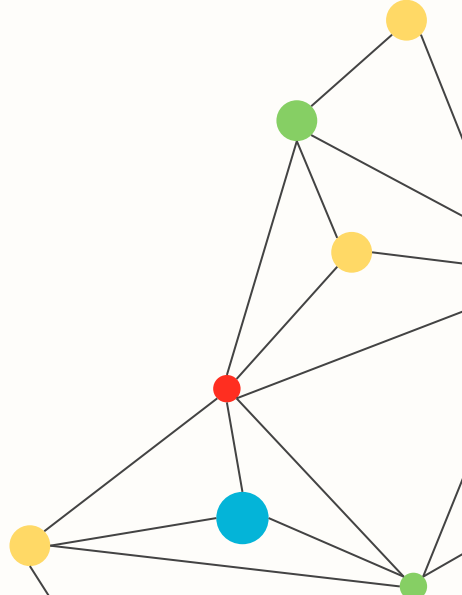
Results – Recommender System Integration

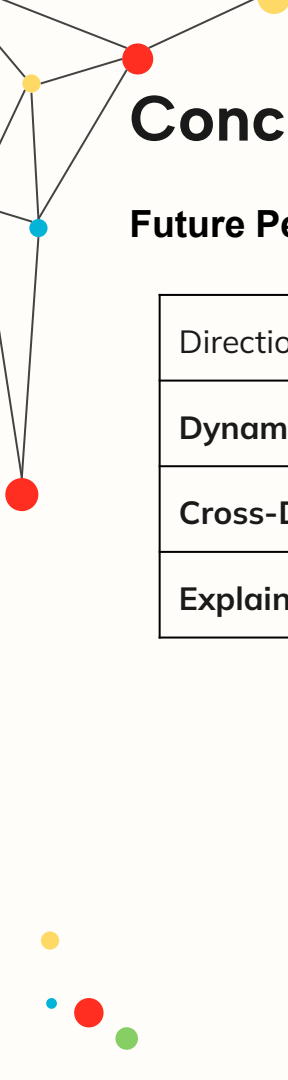
Performance Summary





WEB APPLICATION

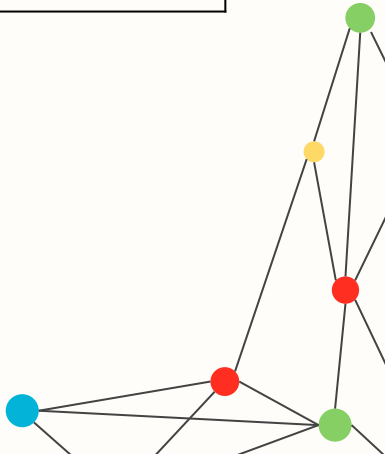


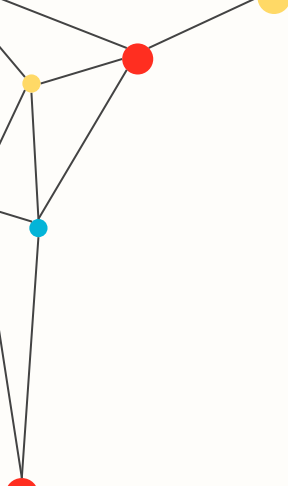


Conclusion & Future Perspectives

Future Perspectives

Direction	Goal
Dynamic Detection	Real-time, incremental community updates
Cross-Domain	Validate on movies, music, social media
Explainability	User trust through community-based explanations





**Thank you for your
attention**

